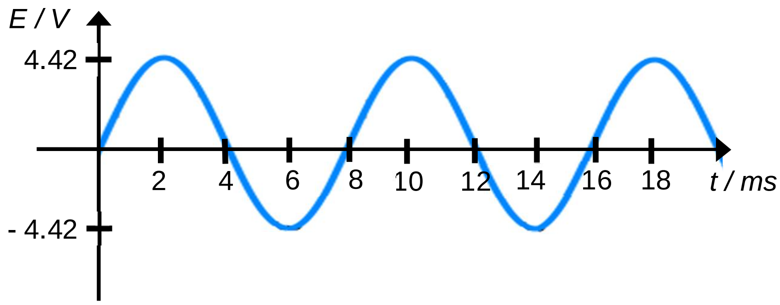
8	**(a)**	**(i)**	The **<u>product</u>** of the **<u>magnetic flux density</u>** **<u>normal</u>** to the surface and the **<u>area of the surface</u>**.	**B1**
		(ii)	$\phi = BA =$ 0.025×0.150×0.060 $\phi =$ 2.25 x 10^{-4} Wb (also accept -2.25 x 10^{-4})	**A1**
	(b)	**(i)**	As the sinusoidal magnetic flux density passing through the coil is changing, the **<u>magnetic flux linkage</u>** through the coil is **<u>changing periodically</u>**. **<u>By Faraday's law</u>**, an **<u>e.m.f. is induced</u>** or generated in the coil. At times, the magnetic flux density is increasing while at other times it is decreasing. The **<u>polarity of the induced e.m.f. would be different</u>** when **<u>magnetic flux density increases</u>** as **<u>compared</u>** to the polarity of the e.m.f. as **<u>magnetic flux density decreases</u>**, thus generating an alternating current.	**B1** **B1** **B1**
		(ii)	From Fig. 8.2, the magnetic flux density is given by, $B = B_0 \cos\left(\frac{2\pi}{T}t\right)$ By Faraday's law and Lenz' Law, the e.m.f. is given by $E = -\frac{d\Phi}{dt}$, where $\mathbf{\Phi}$ is the magnetic flux linkage. $E = -\frac{dNBA}{dt}$ $E = -NA\frac{dB}{dt} = -NA\frac{d}{dt}B_0 \cos\left(\frac{2\pi}{T}t\right)$ Therefore, $E = NAB_0 \frac{2\pi}{T} \sin\left(\frac{2\pi}{T}t\right)$	**M1** **M1** **A0**
		(iii)	$E = NAB_0 \frac{2\pi}{T} \sin\left(\frac{2\pi}{T}t\right)$ Since the maximum magnitude of induced e.m.f. is when $\sin\left(\frac{2\pi}{T}t\right) = 1$, $E_{max} = NAB_0 \frac{2\pi}{T} = 25 \times 0.150 \times 0.060 \times 0.025 \times \frac{2\pi}{0.008}$ = 4.42 V	**C1** **A1**

		(iii)	 sine function max E and period	A1 A1
	(c)	(i)	Root-mean-square voltage of input, $$V_{rms} = \frac{V_0}{\sqrt{2}}$$ $$V_{rms} = \frac{4.42}{\sqrt{2}} = V_P$$ $$\frac{N_S}{N_P} = \frac{V_S}{V_P}$$ No. of turns in sec coil, $$N_S = \frac{V_S}{V_P} N_P = \frac{240}{\frac{4.42}{\sqrt{2}}} \times 15 = \frac{240\sqrt{2}}{4.42} \times 15$$ $$N_S = 1150$$	C1 M1 A1
		(ii)	For efficiency, $\frac{\text{Power output}}{\text{Power input}} \times 100\%$ $= \frac{\text{Power in secondary coil}}{\text{Power in primary coil}} \times 100\%$ $= \frac{0.25 \times 240}{12I} \times 100\% = 81\%$ Therefore, $I = 6.17\text{A}$	C1 A1
	(d)	(i)	can change (output) voltage efficiently or to suit different consumers/appliances by using transformers	B1 B1
		(ii)	for same power, current is smaller	B1

			less heating in cables/wires or thinner cables possible or less voltage loss in cables	**B1**